

Safety in Transportation

While you are driving on a secondary highway, you glance into your rearview mirror. A 16-wheel transport truck has come up behind you and intends to pass.

You are driving at the posted speed limit. It would not be safe to pull off to the side at this point in the road.

You take another look in your rearview mirror to see if the truck has backed off. While you are checking, a white-tailed deer moves onto the road in front of you! There is no time to stop.

If you were the driver, what would you do in the situation described above? How could either you or the driver of the transport truck have prevented the situation?

In this unit you will explore speed and momentum. A knowledge of these will help you understand what happens during collisions. You will also learn how various safety features

- reduce the chances of having a collision, and
- protect drivers and passengers during collisions and sudden stops.



A Need for Safety

Getting Ready...

- What gender and age groups tend to have the most motor vehicle collisions?
- What are the physical, emotional, and legal consequences of a motor vehicle collision?
- How do alcohol and other chemicals affect one's ability to drive safely?



Where do you think you are more likely to have a collision — on a crowded road or a deserted one? Think of reasons for both.



It is 8:30. You were supposed to be at the party at 8:00. The light turns green and you press on the accelerator as you see the cars in front of you start to move. You don't like the song that is playing, so you lean over to change the ...

CRASH!

While your attention was on your iPod, the cars ahead of you stopped to let an animal cross the road. You have just caused a rear-end collision.

How can you avoid such collisions? You might start by learning about the most common driver errors that cause collisions. By avoiding these, you lessen your chances of having one.

In this chapter you will learn about common driver errors. In addition, you will examine transportation data and studies to try to learn where Canadian traffic problems are, and how to make our roads and drivers safer.

You will also work through scenarios to answer question such as:

- What factors lead to traffic accidents?
- Is there a connection between reaction time and motor vehicle collisions?
- What are the most common injuries resulting from collisions?
- How are vehicles and roads designed to keep drivers safe?

13.1 Analyzing the Danger

READING check

Explain how likely you are to die in a traffic accident than from other causes.

An automobile collision is much more serious than falling off a bicycle or down the stairs. The injuries from vehicle collisions range from scratches and bruising to serious internal damage causing death. Many injuries are severe and long-lasting. Victims may break bones, suffer burns, or receive brain damage.

Think it will happen to everyone but you?

Think again. You are 50 times more likely to die in a road collision than you are to die in an airplane mishap.

Do you ever worry about getting murdered? In Canada, you are five times more likely to die in a traffic accident than to be murdered. Your chances of being injured in an automobile accident are dozens of times greater than that.

The sad thing about these high numbers is that **almost all vehicle collisions are preventable!** Nearly all of the injuries and deaths that occur on Canada's roads each year need not have happened.

In this section you will learn about the chances of dying on the roads compared to dying of other causes. You will also consider which modes of transportation are the most dangerous.



Figure 13.1 Collisions result in more than financial loss. Survivors of crashes may have severe injuries.

Find Out **ACTIVITY**

What Are the Most Common Causes of Death?

This table provides **statistics** about Canadians who died during the period from 1993 to 1997. Statistics are numerical facts about people, transportation, diseases, and other items that are gathered and analyzed in an effort to help people improve their life by avoiding accidents or reducing their chances of getting a disease. Statistics can be used to learn about why problems happen. **These statistics show the number of people who died of each cause per 100 000 of population.**

Table 1 Percentage of Driver Injuries and Fatalities, by Age Group (2000)

Age Group	Serious Injuries	Fatalities	Number of Licensed Drivers (thousands)
15–19	11.0	9.8	1088
20–24	15.5	13.3	1693
25–34	20.7	18.6	3875
35–44	19.7	17.9	4896
45–54	15.0	15.8	4067
55–64	7.7	9.3	2470
65+	9.4	14.8	2497
Totals*	99.0	99.5	

SKILLCHECK

Initiating and Planning

Performing and Recording

✶ Analyzing and Interpreting

Communication and Teamwork

Table 2 Canadian Causes of Death

Cause	1993	1994	1995	1996	1997
cancer	310	311	306	306	297
diabetes	17	17	18	17	17
heart and blood system diseases	252	243	238	233	225
respiratory diseases	61	60	60	60	60
chronic liver diseases	8	7	7	7	6
infant death and fatal birth defects	8	8	8	8	8
injuries and poisonings	67	64	65	63	60
motor vehicle accidents	13	11	11	11	10
suicide	13	13	14	13	12
homicide	2	2	2	2	2
other causes	112	114	116	114	120

*per 100 000 of population

What Did You Find Out?

- (a) Using the information from Table 1, compare the number of drivers to the number of fatalities and injuries for each age group. What age of driver appears to have the highest and lowest percentage of serious injuries? of fatalities?
 - (b) Suggest reasons for each.
- (a) Use two bar graphs to plot the data for Table 1.
 - On the first graph, plot the data about serious injuries and fatalities according to age group. Use different colours of bars for each age group.
 - On the second graph, plot the number of licensed drivers by age group.
 - (b) Using this information, what group of drivers should have the highest and lowest insurance rates?
- From Table 2, list, in order, the four most common causes of death for Canadians.
 - On Table 2, which causes of death appear to be decreasing overall? Give a possible reason for each decrease.
 - (a) Which causes of death can you take action to prevent?
 - List some prevention strategies for each of these.
 - Table 2 statistics represent people of all ages.
 - Which causes of death would likely be more common for elderly people?
 - Which causes of death would likely be more common among teenagers?

Injuries

Each year, more than 200 000 Canadians are injured in motor vehicle crashes. Their injuries include:

- cuts, bruises, and broken bones;
- whiplash;
- serious burns;
- internal injuries; and
- head and spinal injuries that may lead to blindness, paralysis, or brain damage.

Each of these injuries has a cost in pain and suffering. Most could have been prevented.

DidYouKnow?

More vehicle collisions happen on Friday than on any other day of the week.

READING Check

List four types of injuries that could result from vehicle accidents.

13.2 Reaction Time

The time it takes you to recognize a problem and act on it is called **reaction time**. In the chapter opener, you were unable to stop because you had not been paying attention. Let's rewind the situation from the chapter opener. Pretend that you did not lean down to adjust the radio station. Could you have stopped in time?

Even if you are alert and react very quickly, there are many factors that affect a driver's ability to stop. The road may have been icy. Your car might have had bald tires or poor brakes. As a result, you might have slid into the car in front of you even though you had been paying attention.

Some drivers have a habit of following too closely. These drivers do not always have enough room to stop if the car in front stops suddenly. Look at figure 13.2. What could happen to the second car if one of the horses bolts and runs across the road in front of the first car? What else might happen?



Figure 13.2 What suggestions do you have for these drivers?



What is reaction time?
Why is it important to drivers?

Reaction Time and Distractions

Anything that takes your attention away from what you are supposed to be doing is called a **distraction**. You cannot really stop distractions from occurring outside your car. However, you can control the distractions *inside* your vehicle.

Are cellular phones really a problem? Statistics suggest they are. Dialling or having a conversation on a cell phone can slow reaction time. During some conversations, reaction time can be slowed by up to three times. For example, a driver who usually reacts in two seconds may take six seconds to react. This is why many provinces have laws banning use of a cellular phone while driving.



Explain how distractions might slow reaction time.

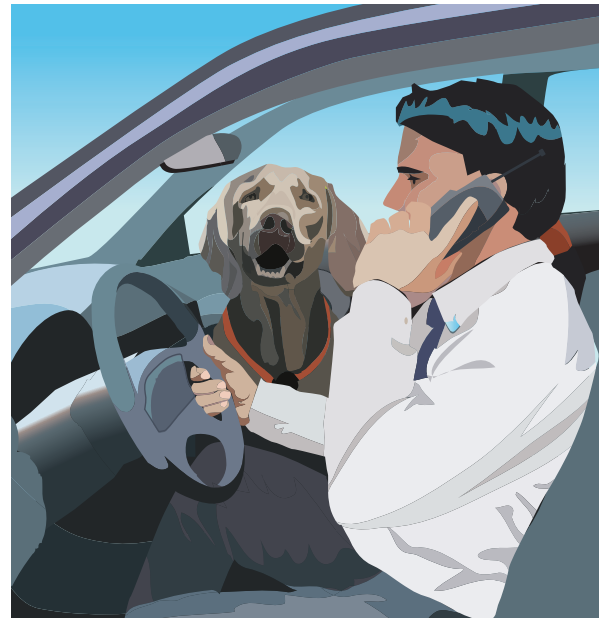


Figure 13.3 Your attention cannot be split between driving and other tasks without creating a danger. Having conversations or tending to children or pets while driving can be very distracting. What traffic law(s) relate to driving distractions?



Figure 13.4 Driving when you are tired can be deadly because fatigue slows reaction time. If you find yourself yawning or blinking, pull over in a safe place and have a nap.

Fatigue and Reaction Time

Fifteen percent of drivers surveyed said they have actually fallen asleep while driving! Sleepy drivers are a serious danger to themselves and others because they cannot pay enough attention to the road and traffic demands. Like drunk drivers, they may

- run off the road;
- roll their vehicle into a ditch; or
- cross the centre line and drive into oncoming traffic, possibly killing other people and themselves.

Some people think they can drink coffee to stay awake, because coffee contains caffeine, which is a **stimulant**. Stimulants increase heart rate and **speed up reaction time**. However, when the effects wear off, the user is more tired than before — and therefore has a slower reaction time.

No amount of caffeine, loud music, cold air, or pinching yourself will keep you awake. Sleep is the only cure. Pull over to a safe parking place and take a nap.

Reaction Time and Drugs

This morning, you woke up feeling sick, so you took a cold remedy before coming to school. Your nose is not running any more, but your head feels woozy. Believe it or not, you are **impaired**. This means you are unable to function normally. Alcohol, prescription drugs, and non-prescription drugs can affect the human body in ways that make it dangerous for a user to drive.

Alcohol is absorbed from the stomach and the small intestine directly into the bloodstream. The human body recognizes alcohol as a poison. The liver produces enzymes that change it into other, safer chemicals. However, once someone has consumed one drink, there is too much alcohol for the liver to deal with all of it at once. The alcohol will move through the blood and have toxic effects.

Alcohol is a **depressant**. This means that it slows reactions, lowers co-ordination, and makes the drinker sleepy when it reaches the brain tissue.

READING check

Explain why alcohol slows reaction time.



Statistics show that many teenagers die while driving at night and with their friends in the car. For this reason, many provinces now have a multi-step driver's licence policy. First you get your learner's licence — you can drive only when you have another, experienced driver in the car with you. Second, you get a licence that allows you to drive under certain restrictions. For example, you might be allowed to drive only during the day or with a certain number of passengers.

In provinces where this graduated licencing system is used, inexperienced drivers have been getting into fewer accidents.

Blood Alcohol Concentration

How do people know whether or not they have had too much alcohol and should not drive? Courts can use a **BAC** (blood alcohol concentration) reading. BAC refers to the number of milligrams of alcohol in a millilitre of blood. If a person's BAC is 0.08, this means that there are 0.08 mg of alcohol in each mL of the person's blood.

Each individual's BAC depends on the number of drinks consumed and body weight. One **drink** is defined as 45 mL of hard liquor (40%), 150 mL of wine (11%), or 350 mL of beer (4%).

How much any individual can drink depends on body weight. Two drinks, one after the other, will put anyone weighing 45 kg or less over the legal limit. A person who weighs 110 kg may be able to have four or five drinks, one after the other, before reaching the legal limit.

BAC does not take into account the amount of food or sleep that a person has had, or the rate at which a person drinks. Generally, having one drink per hour at a party is considered safe for driving. But someone who is tired or hungry can become impaired much more quickly.

BAC also does not take years of drinking experience into account. An older, heavy drinker may have a BAC of 0.8 (10 times higher than the legal limit) and not show any signs of impairment. A younger or smaller person can often have a BAC well below the legal limit and be too impaired to drive safely.



DidYouKnow?

In 1999, 23 percent of all drivers involved in fatal crashes in Alberta had consumed alcohol before the crash.

Figure 13.6 Police measure BAC using a breathalyzer. The law in many provinces says that people with a BAC of 0.08 or over are too impaired to drive.

Blood Alcohol and the Law

What does the law say about drinking and driving? **In Canada, a BAC level of 0.08 is generally considered the legal limit.** However, a police officer can charge drivers who have *any* alcohol on their breath when they are pulled over. The charge is **impaired driving**.

Drivers are considered to be impaired when their reactions are slower than normal. This includes slurred or slow speech and being unable to walk in a straight line.

READING check

What is the legal blood alcohol content for driving in Canada?

DidYouKnow?

Many bars now offer devices that test BAC, so that drinkers can test their alcohol levels before they walk out the door. A BAC level is not the only or the best measure of impairment. People who feel dizzy, giggly, wobbly, or at all unsure of their abilities, should not drive. It is better for such people to stay where they are, and start drinking water or soft drinks instead of alcohol. Many bars also provide free non-alcoholic drinks to designated drivers.

How Soon Can a Drinker Drive?

BAC reports the number of milligrams of alcohol present in 100 mL of blood. Generally, the human body can eliminate alcohol at a rate of 0.015 mg/mL every hour.

Sample Problem

A male who weighs 70 kg and has four quick drinks at a party will have a BAC of about 0.115. If the alcohol leaves the body at a rate of 0.015 BAC units per hour, how many hours will it take until this person is legally able to drive?

Steps in Solving It

1. *What do you know?*
BAC rate is 0.115
Alcohol leaves body at 0.015 BAC units per hour
Legal BAC limit is 0.08
2. *What do you want to find out?*
How long will it take for the drinker to be able to drive legally?
3. *How far over the legal limit is the drinker?*
To find out, subtract the legal limit from the BAC rating.

$$\begin{array}{r} 0.115 \text{ BAC} \\ - 0.080 \text{ BAC} \\ \hline 0.035 \text{ BAC} \end{array}$$

The drinker's BAC is 0.035 units over the limit.

4. *How long will it take this drinker to get down to the legal limit?*

The body gets rid of 0.015 BAC units per hour. You need to calculate the number of 0.015 units there are in 0.035.

$$\frac{0.035 \text{ BAC}}{0.015 \text{ BAC/h}} = 2.33\text{h}$$

If this drinker does not have anything else to drink, it will take slightly longer than 2.3 hours before the person can legally drive.

Solve These

Use the example above to help you solve the following problems. Be sure to show all of your calculations.

1. A 50 kg female who has two drinks will have a BAC level of 0.09. How long will it be until she will be able to drive?
2. Joby drank some whisky with his friends after winning a football game. He weighs 100 kg. His BAC level was 0.117 after six drinks. How many hours must he wait before legally being able to drive?

Key Terms

reaction time
distraction
stimulant
impaired
depressant
BAC
drink
impaired driving

Check Your Understanding

1. Name four things that would increase your reaction time.
2. Alcohol is a depressant. How do depressants affect the human body and reaction time?
3. Why do people drink stimulants like caffeine while driving? How effective is this strategy?
4. Name two reasons why talking on a cellular phone while driving is a dangerous practice.
5. (a) Can alcohol cause impairment before a person is over the legal limit?
(b) Can someone who is under the legal limit for BAC be arrested for impaired driving? Explain.
6. Is talking to a passenger or drinking a soda while driving a wise practice? Explain your reasoning.

13.3 Collision Injuries

Collisions cause damage to more than cars. People involved in a rear-end accident like the one described in the chapter opener may have serious physical injuries. Some may also have emotional problems as a result of the accident.

The physical and emotional stress that results from an accident is referred to as **trauma**. Physical trauma can range from cuts and scrapes to more serious and lasting injuries like broken bones, whiplash, and brain damage.

Some of these injuries heal on their own, while others do not. Physical therapy or Chiropractic care may help to correct some damage. Some injuries cannot be repaired. These include injuries that involve major brain damage, require amputation, or involve the spinal cord.

Physical traumas are often visible. Emotional traumas are not. Some people may feel fearful of all driving after an accident. Others experience emotional trauma associated with grief or regret after an automobile collision that caused serious injury or death. Emotional counselling can assist people with such trauma.



Figure 13.8 Collisions result in more than financial loss. People may spend years trying to walk again after a serious accident.

READING check

What is trauma?
Give three examples.

Whiplash

Whiplash occurs when the muscles and ligaments of a person's neck are sprained. It is a common injury for the people in the front car of a rear-end collision.

Seat belts keep most of the body restrained. Properly adjusted **headrests prevent the head from flying backward beyond the back of the seat**. If the headrests are not adjusted, however, the head may fly forward and then backward during a collision. Since heads have a mass of about 4.5 kg, this movement can over-stretch the neck muscles.

Whiplash is treated in a variety of ways. Chiropractic or massage therapy can be helpful. Some patients wear a neck brace for support while the muscles and ligaments heal.

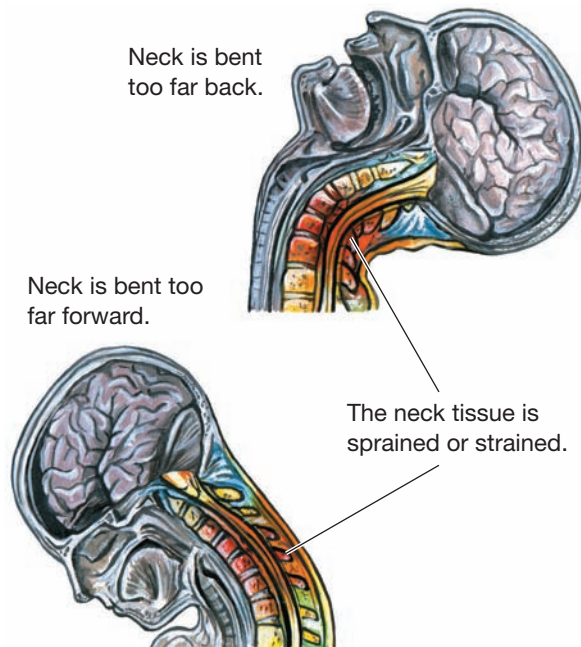


Figure 13.9 Whiplash occurs when the muscles and ligaments of the neck are stretched too far or compressed too much.

READING check

How does a headrest help prevent whiplash?
Draw a diagram to show how it might help.

Grief

How would you feel if you were driving carelessly and got into a serious accident? How would you feel if your passengers died? How would you feel if you survived, but all the people in the other car were killed?

It can take a long time to get over the feelings of regret and sadness we call **grief**. Sometimes, people feel unable to talk about their feelings, or do not allow themselves a long enough adjustment time to get used to their grief. Talking to a professional who understands sadness helps many people deal with this type of emotional pain.

Find Out **ACTIVITY**

What Happens after an Accident?

Many people deal with accident victims, both at the scene of the accident and afterwards. These include:

- police officers and Emergency Medical Technicians (EMTs) who attend the accident itself;
- physiotherapists, chiropractors, and occupational therapists who deal with physical traumas;
- psychotherapists, who help people get over emotional traumas, such as the painful feelings associated with an accident, injuries, or death; and
- members of community groups such as MADD (Mothers Against Drunk Driving) and SADD (Students Against Drunk Driving).

Once the major injuries have been dealt with, these people provide encouragement and therapy that helps the patient return to a normal life.

Therapy is any kind of treatment that does not require surgery or drugs.

In this activity, you will interview someone who works with accident victims.

What to Do



Choose a profession or organization you would like to learn more about. *Use the internet to conduct your research.*

What Did You Find Out?

1. What types of trauma do those involved in motor vehicle collision experience? What role does therapy play in the healing process?

SKILLCHECK

☀ Initiating and Planning

Performing and Recording

Analyzing and Interpreting

☀ Communication and Teamwork

Key Terms

trauma
whiplash
grief
therapy

Check Your Understanding

1. Name two types of trauma. Explain how each type of trauma is treated.
2. Name two types of injuries that can be healed. Name two types of injuries that cannot be healed.

13.4 Protection and Prevention

Safety features are built into vehicles and roads to protect you in case of a road accident. As well, there are laws concerning proper driving procedures and commercial campaigns designed to make us aware of specific traffic dangers.

Safe Vehicles

In Alberta, the law says that every person in a car must wear a seat belt. Small children and babies must be restrained using specially designed seats. Does this law keep you safe?

The statistics say yes. In 1999, people who did not wear seat belts were three times as likely to be injured as those who did wear them.

Seat belts work constantly to restrain passengers. Other safety features are designed to work only when a collision happens.

- **Air bags** inflate when a collision occurs. They cushion drivers and passengers against impact.
- **Anti-lock braking systems (ABS)** provide better stopping ability in slippery conditions.
- **Crumple zones** are on the outer parts of the car, near the bumpers. When these zones crumple, they take the energy of the collision so the passengers feel less effect.
- **Side impact beams** are steel beams in the doors. They prevent the doors from caving in. This protects the people inside the vehicle.

Vehicle Maintenance

Seat belts, air bags, and brakes protect you only when they are working. **Maintenance** or continual care makes sure they continue to do so. Regular maintenance is important for all safety features on your vehicle. These include:

- **tires** — Worn tires, with little or no tread, skid easily. This affects the time it takes for your vehicle to stop.
- **steering** — Taking care of the steering mechanism and all the parts connected to it ensures you can move the car where you want, when you need to.
- **windshield wipers** — These keep your windshield clear so you can see the road ahead. This allows you to identify possible emergencies and increase your chances of stopping when needed.

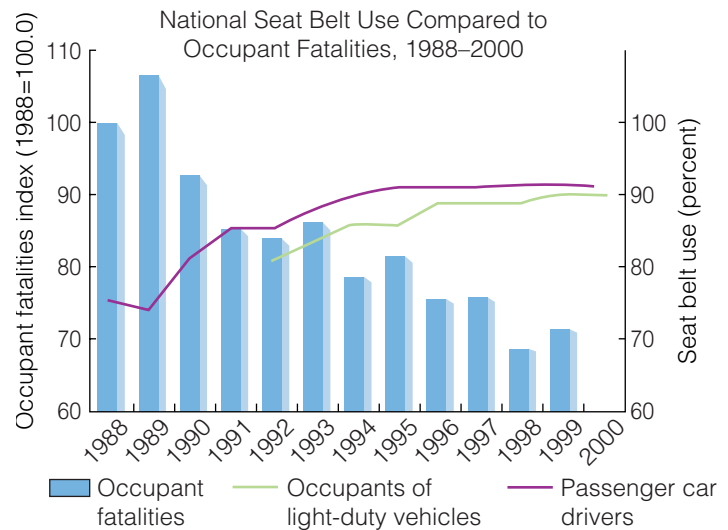


Figure 13.10 What has happened to the number of fatalities since seat belt laws were passed in 1987?

READING check ✓

List the features in modern vehicles that either work to prevent accidents or protect the driver and passenger during an accident. Explain why each should be maintained.

Safe Roads

Roads are designed to be as safe as possible. Most roadways have the safety features shown here.



Figure 13.11 Rumble strips remind people to slow down before intersections or in school zones.



Figure 13.12 Guardrails may protect cars from falling over steep edges.



Figure 13.13 Flashing yellow lights warn of pedestrian walkways or upcoming intersections.



Figure 13.14 Breakaway light poles can fall apart on impact to reduce the force of a collision.

READING check

List three safety devices used on roadways to reduce the impact if struck by a motor vehicle.

Check Your Understanding

1. Which vehicle safety feature do you think is the most important? Explain your answer, using statistics or evidence if possible.
2. Name two forms of maintenance that should be performed frequently on your vehicle. Why are these important?
3. What safety features are built into roads?

Key Terms

maintenance

Getting
Ready...

- How do police use radar equipment?
- What do you need to know in order to graph and calculate speed?
- How far will a vehicle keep travelling after you hit the brakes?



Thrust SSC holds the World Land Speed Record, set on 15 October 1997, when it achieved a speed of 1,228 km/h (763 mph) and became the first car to officially break the sound barrier.

Read the captions for the photographs shown on these two pages. Did you know that peregrine falcons can dive almost as fast as a speeding race car — and a lot faster than most vehicles can go?

Any object that is moving has speed. Some, such as a glacier that might move one metre per year, move very slowly. Others, like particles of light that move 300 000 000 metres per second, travel at great speed!

It is not always easy to judge speed, as you will learn in the Starting Point activity. Astronauts on a space shuttle might not feel like they are moving at all. Yet they are

travelling about a thousand times faster than a person on a roller coaster at an amusement park.

You can see why a mathematical definition of speed and a way of measuring it are important. Human senses are not always accurate enough for safety.

The purpose of this chapter is to answer the question, “What is speed?”

You will use and analyze graphs, work with equations, and learn how one form of police technology that measures speed works. Finally, you will learn how speed affects the stopping distance of objects and how to adjust your driving to handle this.

14.1 Defining Velocity

READING check

Define “velocity.” What two factors are needed in order to determine velocity?

Car speedometers use the two factors scientists use to determine speed. These are distance and time.

- **Distance** is a change in position.
- Change in position occurs over a period of **time**.

In this unit, we will use the term velocity to mean speed. **Velocity is the distance travelled during a specific time interval.** For example, the maximum legal speed on some Alberta highways is 110 km/h. In towns, it is half that — 50 or 60 km/h.

Vehicles travelling at 100 km/h cover a distance of 100 km in one hour. Those going at 50 km/h cover half that distance in an hour.

Notice that all explanations of velocity include units — km/h for vehicles, perhaps m/s for an ant or snail.

As long as a description of velocity includes units, you can get a fairly good idea of how fast something is going. What units would you use to describe the velocities shown in Figures 14.1 to 14.4?

Figure 14.1



Figure 14.2

Table 14.1 Chart of Comparative Top Velocities

Object	Velocity (km/h)	Velocity (m/s)
light	1 080 000 000 km/h	300 000 000 m/s
space shuttle	28 000 km/h	7778 m/s
cheetah	115 km/h	32 m/s
antelope	88 km/h	24 m/s
black tuna	80 km/h	22 m/s
salmon	60 km/h	17 m/s
hawk moth	53 km/h	15 m/s
human	37 km/h	10 m/s
stegosaurus	7 km/h	2 m/s
garden slug	0.05 km/h	0.01 m/s

km/h = kilometres per hour
m/s = metres per second

Figure 14.3

Figure 14.4

Calculating Velocity

Radar guns calculate velocity almost instantly. The radar gun shoots two laser beams at a vehicle, and then the beams bounce back to a sensor. The two beams are a certain period of time apart. The sensor judges the difference in position during that period of time. A calculator in the radar gun finds the velocity from the distance and time measurements.

READING

check

How does a radar gun calculate a car's velocity?

You can do this yourself using a simple formula.

$$\text{velocity} = \frac{\text{distance}}{\text{time}} \quad \text{or } v = \frac{d}{t}$$

d = distance in kilometres (km) or metres (m)

t = time in hours (h) or seconds (s)

v = velocity in km/h or m/s

Any time you know the distance an object has travelled and the time it has taken to cover the distance, you can calculate the object's average velocity over that period of time by division.

Sample Problem

A snowboarder makes it down a 1000 m run in 60 seconds. What is her velocity?



4. Show the velocity as distance per time.

$$v = \frac{1000 \text{ m}}{60 \text{ s}} \text{ or } 1000 \text{ m}/60 \text{ s}$$

5. Divide the time units into the distance units.

$$v = \frac{1000 \text{ m}}{60 \text{ s}} = 17 \text{ m/s}$$

The snowboarder moves at approximately 17 m/s.

Steps in Solving It

1. What do you know?

$$d = 1000 \text{ m}$$

$$t = 60 \text{ s}$$

$$v = ?$$

2. What do you want to find out?

How fast is the snowboarder moving?

3. How do you find out?

$$v = \frac{d}{t}$$

Solve These

Use the examples above to help you solve the following problems. Be sure to show all of your calculations.

1. To get home you drive 25 km in half an hour. At what velocity are you travelling?
2. At what velocity would you have to drive to travel 20 km in 15 minutes (1/4 h or 0.25 h)?

14.2 Graphing Velocity



Figure 14.5

They say a picture is worth a thousand words. A graph is just that — a picture. It is difficult to make sense of a table full of numbers that show distance and time. If you plot these numbers into points on a graph, however, you can see all the information at once.

In this section, you will see and use different graphs involving velocity. These are called distance versus time graphs. For short, we will call them **distance/time graphs**. In distance/time graphs:

- Time is plotted on the x -axis (horizontal).
- Distance is plotted on the y -axis (vertical).
- The line that joins the data points gives information about the velocity.

Distance/Time Graphs

In distance/time graphs like the one in Figure 14.6, the title

- describes the object being studied (lawnmower) and
- what **variables** (changing aspects of the investigation) were measured for the graph (distance and time).

The x -axis and y -axis show

- the units for each variable (Time, measured in seconds, is on the x -axis. Distance, measured in metres, is on the y -axis.)
- the x -axis shows the **manipulated variable**. It is the one variable that is changed by the researcher (in this case, time).
- the y -axis shows the **responding variable**. It is the variable that is measured as it changes as a result of changes to the manipulated variable (in this case, the distance the lawnmower travelled during each time interval).

Finally, the **slope** of the line shows that the lawnmower is moving at a constant velocity.

Try This!

What kinds of graphs have you already used in science? How was each kind of graph used?

READING check

What part of a distance/time graph tells about the velocity?

Distance/Time Graph for a Lawnmower

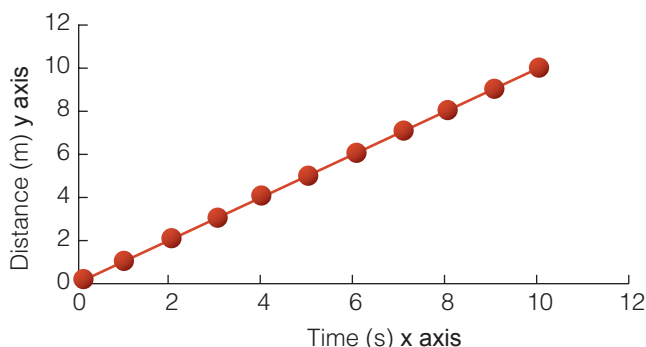


Figure 14.6 What does this graph show?

Constant Velocity, Changing Velocity

Look at the two graphs on this page. In Figure 14.7, a straight line shows a constant velocity. A jagged or curved line, like the one shown in Figure 14.8, shows that the velocity of the object changed over time. The slope of the line shows the velocity of the object.

READING check

During what time interval was the object travelling slowest in Figure 14.8? When was it travelling fastest?

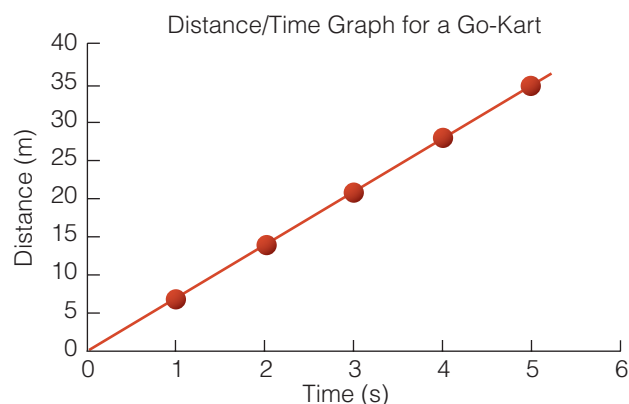


Figure 14.7 The straight line on this distance/time graph means that the vehicle was going at a constant velocity.

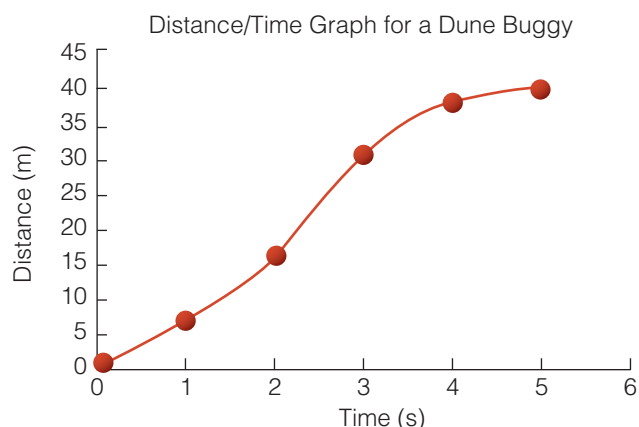


Figure 14.8 The curving line on this distance/time graph means that the vehicle was changing velocity.

Find Out **ACTIVITY**

What Are You Plotting?

The data here were collected by timing a running student for 10 seconds. Ten classmates stood along the track to mark the runner's position. A timer yelled at each second. The nearest classmate marked the runner's position.

What You Need

graph paper
ruler
pencil

What to Do

- Look at the table of data shown here. Determine whether the position numbers change regularly or at different time intervals.
- Predict whether you think these data will result in a straight line.

Position and Time Data	
Time (s)	Position (m)
1	5
2	10
3	15
4	20
5	25
6	30
7	35
8	40
9	45
10	50

- Use the data to create a distance/time graph. Your graph should follow the same format as the one in Figure 14.7.

What Did You Find Out?

- As time goes by, does the runner's position change regularly or in unpredictable ways?
- Does the runner's velocity stay the same or change?
- Predict how the line would look if the runner's position changed by 6 m instead of 5 m during each time interval.
 - Redo the data chart showing these new data. Graph your new data.

SKILLCHECK

Initiating and Planning

- Performing and Recording
- Analyzing and Interpreting
- Communication and Teamwork



Define the terms “rise” and “run.”

Calculating Velocity from a Distance/Time Graph

In Conduct an Investigation 14–B, you made distance/time graphs using data from a data table. You plotted each distance and each time as points or co-ordinates on your graph. You can use such a graph to determine the velocity of an object. The method is shown in the Math Connect below.

Before starting on the Math Connect, study Figures 14.9 and 14.10. Note that the **run** on a distance/time graph shows the time interval between two points. The **rise** shows the distance interval between the same two points.

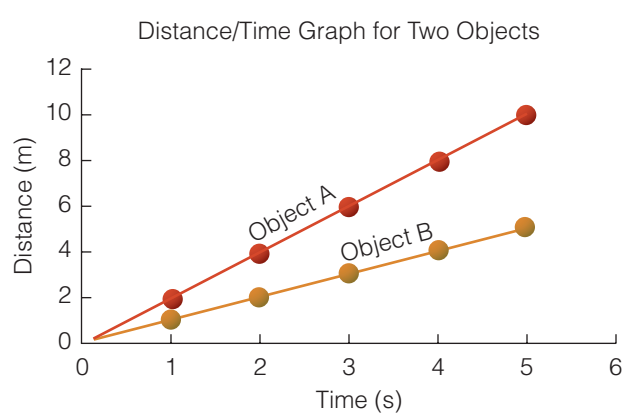


Figure 14.9 Both objects are moving at a constant velocity. Object A is moving faster than object B.

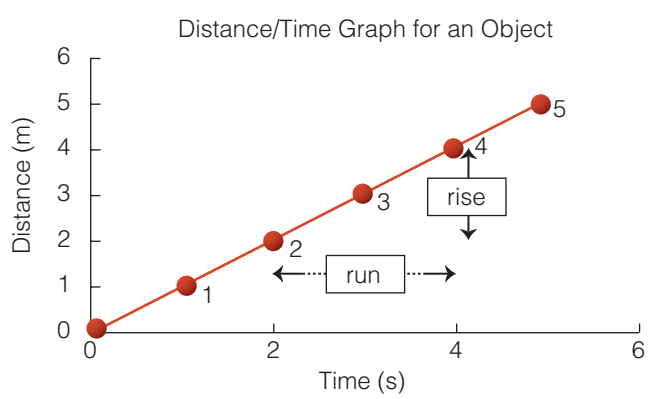


Figure 14.10 Identify the rise and run on this graph.

Math CONNECT

Slope Formula

The line on a distance/time graph represents the velocity of the object being studied. As you see in Figure 14.9, the steeper a line is, the faster the object is moving.

The steepness of the line is referred to as its **slope**. You can use the slope of a line to calculate exactly how fast an object is moving.

To find the slope:

1. Choose two points on the line. Identify the co-ordinates of the two points. For example, mark the number on the x- and y-axis for each point.
2. Find out the rise. This is the distance interval between the two points. To calculate the rise, find the difference between the two y-axis points.
3. Find out the run. This is the time interval between the two points. To calculate the run, find the difference between the two x-axis points.

4. To find the velocity, divide the calculation from Step 2 (rise) by the calculation from Step 3 (run).

$$v = \frac{\text{rise}}{\text{run}} \text{ or } \frac{\text{distance}}{\text{time}}$$

Sample Problem

What is the slope of the distance/time graph for Granny's car?

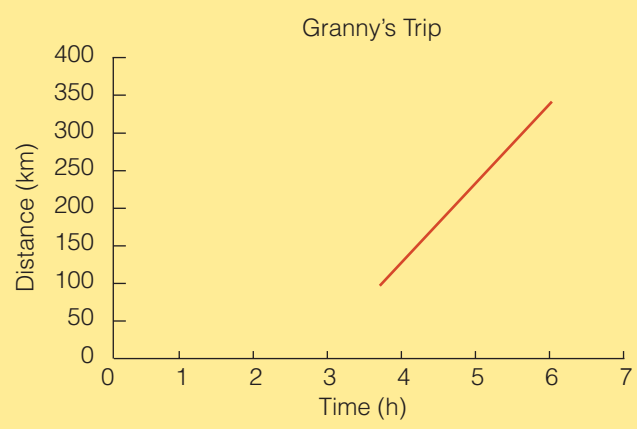


Figure A

Steps in Solving It

1. What do you know?

Two points on the graph are (100, 3.6) and (340, 6.0).

The distance units are in kilometres.

The time units are in hours.

2. What do you want to find out?

How fast is Granny travelling?

3. How do you find out?

i. How far did Granny travel?

Calculate how far Granny travelled between the two points on the graph (distance = points on the x-axis.)

$$d = 340 \text{ km} - 100 \text{ km} = 240 \text{ km}$$

ii. How long did it take Granny to travel this distance?

Calculate how long it took Granny to travel between these two points (time = points on the y-axis.) This will give you a time measurement.

$$t = 6.0 \text{ h} - 3.6 \text{ h} = 2.4 \text{ h}$$

iii. What do you know?

$$d = 240 \text{ km}$$

$$t = 2.4 \text{ hr}$$

$$v = ?$$

$$v = \frac{d}{t} = \frac{240 \text{ km}}{2.4 \text{ h}} = 100 \text{ km/h}$$

4. Show the velocity as distance per time.

Granny is travelling at 100 km/h.

Solve These

Use the example above to help you solve the following problems. Be sure to show all of your calculations.

- Calculate the velocity of the go-kart by calculating the slope of the graph in Figure B. Show all your steps.

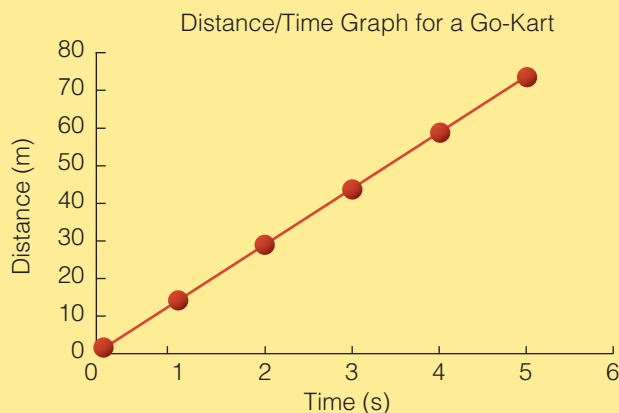


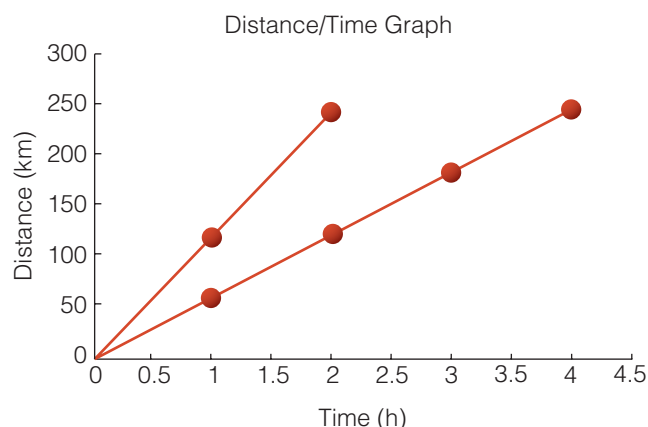
Figure B

Check Your Understanding

- What does the slope of a line on a distance/time graph represent?
- What does the horizontal line on a distance/time graph represent?
- In a distance/time graph, which variable is manipulated? Which variable is responding?
- (a) There are two lines on the distance/time graph shown here. One line is twice as long as the other. What does that represent?
(b) If the slope of one line is twice that of the other one, what does that suggest?

Key Terms

distance/time graphs	run
variables	rise
manipulated variable	slope
responding variable	



14.3 Calculating Distance

READING

check

How is the formula for distance related to the formula for velocity?

Before travelling to Europe, you calculated the number of kilometres you would travel directly east to reach your destination. During the flight, you learned that the actual flightpath took a shortcut by heading north. You learned the airspeed of the plane from the pilots, and spent the rest of your eight-hour flight calculating the length of the shortcut.

To find distance, you multiplied the velocity of the airplane times the amount of time you travelled at that velocity. The formula you used is $\text{distance} = \text{velocity} \times \text{time}$.



Math CONNECT

Using a Distance/Time Graph to Calculate Distance Travelled

You now know how to calculate velocity by using a graph, or by dividing distance by time. The formula you use is $v = \frac{d}{t}$. As you learned in the Math Connect on page 260, the slope on a distance/time graph represents velocity.

How could you use the information on a distance/time graph to calculate the distance travelled?

If you knew the velocity (the slope of the graph) and the time (the run of the graph), you could multiply them to find the distance (the rise of the graph). You can find the distance an object has travelled by multiplying its velocity times the time it travelled.

$$d = vt$$

Sample Problem 1

You decide to get into shape for the football season by riding a stationary bike for 30 minutes (0.5 hours) each day. If you were pedalling at an average speed of 15 km/h, what would the odometer read at the end of one session?

Steps in Solving It

1. *What do you know?*
time = 0.5 h
velocity = 15 km/h
distance = ?
 $d = vt$
2. *What do you want to find out?*
How far you would have travelled.
3. *How do you find out?*
 $d = vt$
 $= 15 \text{ km/h} \times 0.5 \text{ h}$
 $= 7.5 \text{ km}$
4. *Summarize your findings.*
You would have travelled 7.5 km.

Sample Problem 2

Your company asks you to travel from Leduc to Lethbridge in the summertime. You carefully keep to 100 km/h on the secondary highways, and subtract refreshment stops from your time record. When you reach Lethbridge, you calculate it has taken 4 hours and 15 minutes. This is 4.25 h. You are being paid by the km. How many km will your company have to pay you for?

Steps in Solving It

1. *What do you know?*

time = 4.25 h

velocity = 100 km/h

distance = ?

$$d = vt$$

2. *What do you want to find out?*

How far have you travelled?

3. *How do you find out?*

$$d = vt$$

$$= 100 \text{ km/h} \times 4.25 \text{ h}$$

$$= 425 \text{ km}$$

4. *Summarize your findings.*

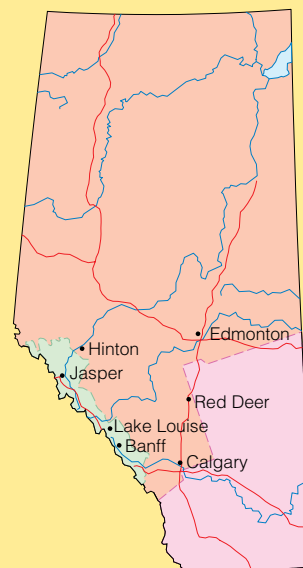
Your company will have to pay you for 450 km.

Solve These

Use the examples above to help you solve the following problems. Be sure to show all of your calculations.

1. What distance will you travel in 2.3 h at an average of 56 km/h?

2. Imagine you are going on a trip with many stops. Calculate and fill in the missing information for the Distance/Time Chart for Alberta.



**Distance/Time Chart
for Alberta**

	Distance	Time	Speed
Edmonton–Red Deer		1.35 h	110 km/h
Red Deer–Calgary		1.33 h	110 km/h
Calgary–Banff	159 km	1.76 h	
Banff–Lake Louise	56 km	0.62 h	
Lake Louise–Jasper	289 km	3.2 h	
Jasper–Hinton		0.66 h	100 km/h
Hinton–Edmonton		2.48 h	110 km/h

Check Your Understanding

1. According to the average velocities shown in the log book, how far did the truck travel between 11:30 and 12:00?
2. According to the average velocities shown in the log book, how far did the truck travel between 6:00 A.M. and 7:30 A.M.?
3. A dog trots along behind a cowgirl's horse at 6 km/h. It takes the cowgirl three hours to get back to the ranch after checking the herd, but she takes half an hour of that time for a lunch break. How far did the dog trot?
4. (a) If you travel 60 km/h for two hours, then an average of 30 km/h for half an hour, then 100 km/h for half an hour, how far have you travelled?
(b) Write a scenario to explain where you might have travelled at these velocities.

Mountain Sky Transport	
Daily Kilometre Log	
06:00–09:30 clear roads, headed out from Edmonton for Hinton. Maintained 100 km/h with no stops clear through to Jasper.	
09:30–10:15 had coffee while waiting to be unloaded	
10:15–11:15 clear driving at 100 km/h	
11:15–13:00 bad snowstorm, averaged 60 km/h	
13:00–13:30 lunch stop	
13:30–16:00 back to Edmonton, averaged 80 km/h	

Distance Travelled during Reaction Time

READING check

How do 30 km/h speed limits near schools and playgrounds help save the lives of children?

DidYouKnow?

A car travelling at 50 km/h will travel 42 m in 3 s. At 120 km/h, the same car will travel 100 m in 3 s.

Why are speed limits different for different roads?

In a school or playground zone, there is a much greater chance that a child will jump into traffic. The 30 km/h speed limit is set to ensure that drivers will be able to stop fast enough to save the life of the child.

Do drivers' reaction times change with greater velocities? Probably not. However, the distance travelled during the time required to react will be much greater at higher velocities.

Imagine you were driving 80 km/h down a city street (with a normal limit of 50 km/h). If someone decided to run a red light at the intersection just in front of you, you would have no way of stopping in time.

In Chapter 13, you found out your reaction time. In this section you will see how far a vehicle would travel during that reaction time before coming to a stop. When driving, reaction time is the time it takes between the driver seeing the need to stop and applying the brakes.

READING check

What needs to be added to the stopping distance to give a true picture of how far a vehicle will travel between the time you see a danger and the time the vehicle stops?

Reaction Time and Stopping Distance

Some driver education programs have new drivers do a simple exercise on a closed course. Student drivers drive around the course at 60 km/h. Then, on the signal from the instructor, they hit the brakes. What do you think happens?

The car will travel several metres before stopping. **The distance that a vehicle travels after the driver applies the brakes is called the braking distance.** **Stopping distance** includes the distance travelled during reaction time *plus* the braking distance.

What if you are reacting to a sudden obstacle in the road only 10 metres ahead of you? If you are going 60 km/h, it would be impossible to stop immediately.

Yellow traffic lights are set to last different lengths of time depending on the speed limit of the road. This is to make sure people have enough warning to see the light change, react, *and* stop their vehicle before the light turns red.

Table 14.2 Stopping Distance Based on Velocity

stopping distance = distance travelled during reaction time + braking distance

Velocity	Distance Travelled During Reaction Time		Braking Distance
	Distance Travelled between When Driver Sees Item and Realizes Must React (approximate)	Distance Travelled between Driver Recognizing Need to Respond and Applying Brakes (approximate)	Distance Travelled between When Brakes Applied and When Vehicle Stops (approximate)
32 km/h	6 m	6 m	6 m
48 km/h	9 m	9 m	12 m
64 km/h	12 m	12 m	22 m
80 km/h	15 m	15 m	35 m
96 km/h	18 m	18 m	50 m

Stopping distance includes the three measurements in this table.

1. Calculate the total stopping distance at each velocity by adding the data in each row.
2. Notice the difference between the stopping distance at 32 km/h and at 64 km/h. **When the velocity doubles, how much does the stopping distance increase?**



What is the recommended following distance? How do you check if you are keeping it properly?

Following Distance

Cars that follow too closely behind others are at a high risk of having a collision.

No matter how fast your reaction time is, a vehicle takes a certain time and distance to stop. The faster it is moving, the longer it will take to stop.

For safety, you should keep a following distance of two seconds behind the car in front of you. This means that when the car in front of you passes a road sign, it should be two seconds before you pass the same sign.

Following too closely is against the law. The police can give you a ticket for tailgating.

DidYouKnow?

A car travelling at 50 km/h will take about 15 metres to stop. The same car travelling at 100 km/h will take at least 75 metres to stop.

Check Your Understanding

1. What is the relationship between velocity and stopping distance?
2. What are two reasons for setting a yellow light to last longer?
3. Explain why drivers should adjust their following distance according to road and weather conditions.
4. If you were travelling 100 km/h in an area where the speed limit was 50 km/h, how would this affect your ability to react to what was going on in front of you? Explain.

Key Terms

braking distance
stopping distance
following distance

The Physics of

Getting Ready...

- Why does it take a train longer to stop than a car?
- Why does it take longer for a fast-moving vehicle to stop than for a slow-moving one?
- Why do collisions usually do less damage at lower velocities?
- Why do they say, “the bigger they are, the harder they fall”?



Imagine this: you are driving on a busy one-way street to pick up some medications at the store. You are rested. There is nothing to distract you, and there are no medications or alcohol in your system. Your speed is within the posted limit. The road conditions are great. You are even following at a gap of two seconds!

Then, without warning, a car turns into your lane just ahead. No matter how fast your reaction time, one way or another you are going to hit something. You have a choice between the car that turned in front of you, the van to your left that is going in the same direction as you,

or the row of parking meters to your right. What do you do?

You have three options. Which one will put you in the least amount of danger? That depends on several factors.

- How heavy are the objects?
- At what velocity are the objects moving?
- What direction is their motion?

In this chapter you will learn how mass and direction of the objects involved in a collision affect the severity of the damage. You will also have a look at how the forces act in a collision.

15.1 Momentum and Mass

DidYouKnow?

Isaac Newton studied momentum. His first law of motion states that an object in motion will stay in motion until acted on by an outside force. Does that force need to be greater to stop a train than it needs to be to stop a bicycle?

DidYouKnow?

A supertanker moving at 30 km/h may take 20 minutes and 5 km to come to a stop after the engines are shut off.

READING check

What factors determine the force required to move an object?

off the Wall

For a mosquito to have the same momentum as a transport truck travelling at 100 km/h, the mosquito would have to be moving at 900 billion km/h!



Figure 15.1 Which boat is easier to move or stop?

Imagine a swimmer in front of the above boats. Would the swimmer be able to stop the canoe? How about the tanker ship?

To answer these questions, you have to know a little about momentum and force.

- **Momentum** is a measure of an object's motion and how likely it is that the object will remain in motion.
- A **force** is any push or pull on an object. The size of the force required to move an object depends on how long you apply force, the mass of the object, and how fast you want to make the object go.

Objects that are not moving will not move until they are pushed or pulled with enough force. Objects that are moving will keep moving until enough force is added in an opposite direction to stop them. Objects with a lot of momentum will take more force and longer to stop.

- A swimmer is able to stop a canoe because the swimmer and canoe have a similar mass, and the canoe moves slowly enough that the swimmer can exert a force on the canoe for a long enough time to stop it.
- A swimmer cannot stop a ship because the ship is very heavy. As well, the ship is moving so fast that the person would not have time to exert a small force for long enough to stop the ship before being run over.



Figure 15.2 554 km/h! The magnets in the track of this levitating train actually hold it up in the air. The friction that the train experiences is almost zero! That is how it can reach such super speeds.

Calculating Momentum

Momentum is calculated by multiplying the mass of an object times the velocity it is travelling. Heavier and faster objects have more momentum.

The formula for momentum is:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

When calculating momentum, we usually express:

- mass in kilograms (kg)
- velocity in metres per second (m/s)
- momentum in kilogram metres per second (kg•m/s)



Figure 15.3 Why does this sign show a truck instead of a car?

READING Check

Write the formula for calculating momentum. Show the units for each part of the formula.

Math CONNECT

Calculating Momentum

Use the following steps to help you calculate momentum when you know velocity and mass.

Sample Problem

A 60 kg girl jogs at a pace of 4 m/s. What is her momentum?

Steps in Solving It

1. Write down what you know.

mass = 60 kg
velocity = 4 m/s
momentum = ?

2. What do you want to find out?
What is her momentum?

3. Use the formula.
momentum = mass $\times$ velocity
= 60 kg $\times$ 4 m/s
= 240 kg•m/s

4. State your findings.
The girl's momentum is 240 kg•m/s.

Solve These

Use the example above to help you fill in the missing parts in the chart that follows. Be sure to show all your calculations.

Object	Mass	Velocity	Momentum
adult female fly	0.000 012 kg	5 km/h (1.5 m/s)	
adult female human	70 kg	5 km/h (1.5 m/s)	
adult female on 5 kg bicycle	75 kg	30 km/h (9 m/s)	
adult female and motorcycle	250 kg	50 km/h (15 m/s)	
compact car	750 kg	50 km/h (15 m/s)	
compact car	750 kg	100 km/h (30 m/s)	
transport truck	2500 kg	50 km/h (15 m/s)	
transport truck	2500 kg	100 km/h (30 m/s)	

Slowing Momentum

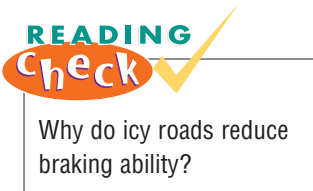
As every cyclist and driver knows, the best way to counteract momentum is by **braking**. Brakes apply a force that stops vehicles by decreasing their momentum.

Brakes work using friction. Friction is the force that two objects exert on each other when they press against each other. Braking uses friction in two ways:

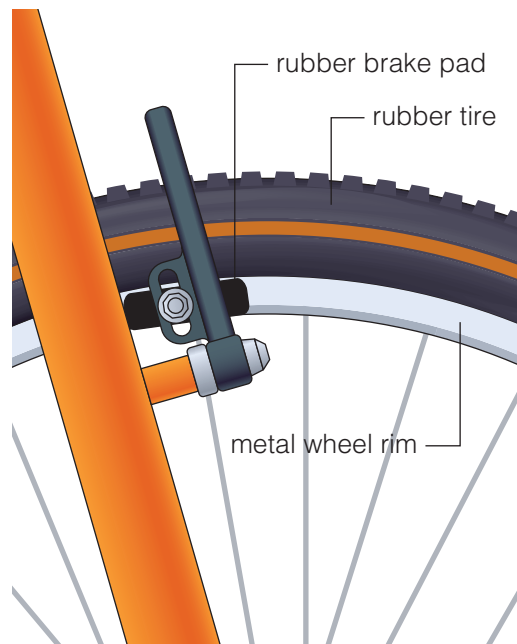
- Applying the brake pedal activates machinery that presses brake pads against the inside of brake drums on the vehicle's wheels. That is one source of friction that slows momentum.
- There is also friction between the vehicle's tires and the surface of the road. During braking, this friction reduces momentum.

Brakes work best in dry conditions when the brake machinery is in good condition. Worn brakes do not work as quickly against momentum. In addition, roads that are slippery due to rain, ice, or snow will reduce the amount of friction and therefore slow the speed of braking.

Friction also works with canoes and kayaks, in which you can stop paddling and gradually glide to a stop.



A. Bicycle Brakes



B. Car Brakes

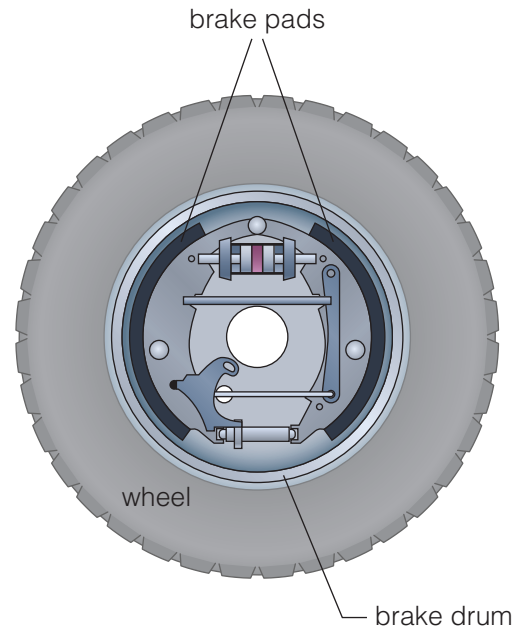


Figure 15.4 Bicycle brakes and car brakes work in essentially the same way — pads press against the wheel until the friction slows the wheels.

Changing Momentum Is Related to Force

DidYouKnow?

Isaac Newton wrote most of the laws about forces.

The unit for force ($\text{kg}\cdot\text{m}/\text{s}^2$) is called a newton (N) in his honour.

READING check

What is an impulse, and how does it affect the momentum and velocity of an object?

As you approach an intersection, you can see the light turning amber. You begin braking, and the force of friction created by your brakes slows your vehicle. Your velocity decreases slowly, as does your momentum (remember $\text{mass} \times \text{velocity} = \text{momentum}$).

Because your body has its own momentum, you need a force to slow *it* as well. The seat belt provides that restraining force. Because you are taking a long time to brake, that force will be small.

Forces, like friction, can be large or small. You can brake hard or gently.

Forces can also either act for a short time or a long time. In our example, you brake gradually for a long time (small force $\times$ long time). If you were to brake hard, the car would stop quickly, and you would feel a great force (large force $\times$ short time).

Either way, the result would be the same. The car would go from a certain velocity (and therefore momentum) to zero velocity. **The combination of the size of the force and the time during which the force acts is called impulse. Impulses change momentum — either increasing or decreasing it. The bigger the impulse is, the bigger the change in momentum will be.**

$$\text{impulse} = \text{force} \times \text{time}$$



Figure 15.5 This sumo wrestler is at rest. If you wanted to move him (give him some momentum), you would have to apply a large force to start momentum.

Measuring Impulse

When working with impulse, you will work with specific measurements. These include:

- **impulse** — This is measured in $\text{kg}\cdot\text{m}/\text{s}$.
- **force** — This is measured in $\text{kg}\cdot\text{m}/\text{s}^2$ or newtons (N).
- **time** — This is measured in seconds (s).

Calculating Impulse

Use the information on measuring impulse and the formula as you work through this Math Connect.

Sample Problem

A car is hit from behind by an identical car with a force of $50\,000\text{ kg}\cdot\text{m/s}^2$. Both cars come to a stop within 2 seconds. What impulse was experienced?

Steps in Solving It

1. Write down what you know.

$$\text{force} = 50\,000\text{ kg}\cdot\text{m/s}^2 \text{ or } 50\,000\text{ N}$$

$$\text{time} = 2\text{ s}$$

$$\text{impulse} = ?$$

2. What do you want to find out?

What is the impulse?

3. Use the formula.

$$\text{impulse} = \text{force} \times \text{time}$$

$$= 50\,000\text{ kg}\cdot\text{m/s}^2 \times 2\text{ s}$$

$$= 100\,000\text{ kg}\cdot\text{m/s}$$

4. State your findings.

The impulse of the object in this example is $100\,000\text{ kg}\cdot\text{m/s}$.

Solve These

Use the example to help you solve the following problems. Be sure to show all your calculations.

1. A football player collides with another player. The force of the collision is $1000\text{ kg}\cdot\text{m/s}^2$. The collision lasts 1 second. What is the impulse?
2. What would be the impulse created by the same force of tackle ($1000\text{ kg}\cdot\text{m/s}^2$), but this time lasting for 2 seconds?

Impulse and Force

Compare your answers to problems 1 and 2, in the above Math Connect. What happens to the impulse if the force is applied for a longer period of time?

If you did the problems correctly, **you should notice that the impulse (or change in momentum) increases as force is applied over a longer period of time.** Impulse also increases if a larger force is applied.

- A force of 500 N over 2 seconds will have a greater impulse than a force of 50 N over 2 seconds.
- A force of 50 N for 5 seconds will have a greater impulse than a force of 50 N for 2 seconds.

A large impulse can change momentum a great deal. This means that:

- **You can change momentum by applying a greater force.**
- **You can change momentum by applying a force for a longer period of time.**

Which method of changing momentum would you want to use to avoid a collision? Which method would you use if you wanted to move a house without damaging it?

READING
check

List three different ways to change momentum.

15.3 Changing Momentum Quickly

Drivers frequently drive at 100 km/h (28 m/s) and stop safely without injuring themselves or damaging their vehicles.

Imagine the force you would feel if your vehicle went from 100 km/h (28 m/s) to a complete stop in less than one second! This is what occurs in a crash, and the forces involved are great enough to crumple metal, shatter glass, and splinter bones.

The difference in the two stops is the time involved. Fast stops require a large impulse that results in an immediate change in momentum. In order to do this, a large amount of force is required. This force is what causes injury and property damage during collisions.

The Math Connect on page 278 shows the difference in impulse between these two examples.

In both examples, note that several factors were the same.

- The change in velocity was the same.
- The change in momentum was the same.
- The impulse was the same.

Only the amount of time it took to achieve a change in velocity was different. It is this time difference that makes the difference between a small force and a great one. This difference in force may also be the difference between life and death for a vehicle's passengers. Figure 15.7 shows why.

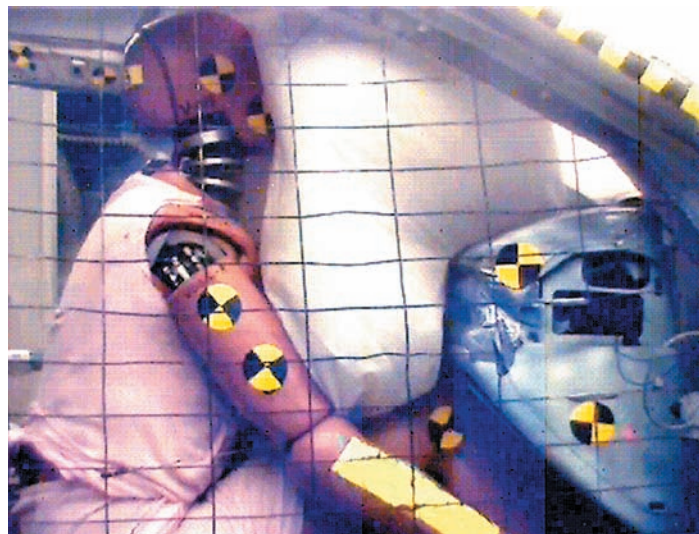
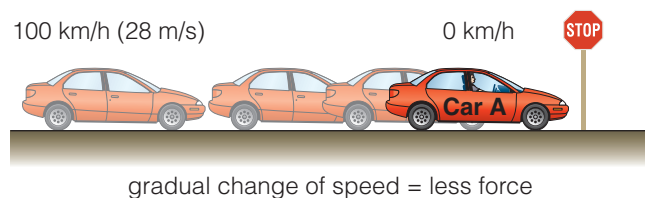


Figure 15.6 Seat belts, air bags, and other safety features help to slow passenger momentum, even when the car experiences a large impulse. The car and passenger experience the same impulse during this crash. The car experiences the impulse in a very short time (large force). The passenger, however, feels it over a longer time as the seat belt and air bag slow momentum gradually. More time = less force.

READING
Check ✓

How is the force of a collision related to the time it takes for a change in momentum?

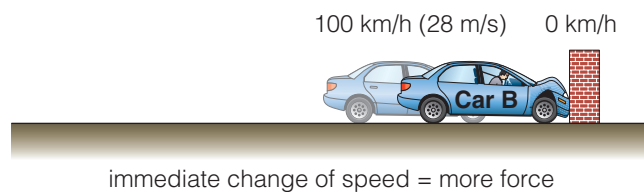


Figure 15.7 These two cars were travelling at the same velocity. Car A braked and stopped over 25 seconds. Car B ran into a brick wall and stopped in 1 s. Which driver will experience a greater force? Explain. Use the information in the Math Connect on the next page to prove your point.

Calculating Force

When an impulse occurs, momentum changes. Usually, the mass stays the same and the velocity changes.

The amount of time it takes to change velocity determines the force involved. For example, it takes more force to stop a car going 100 km/h in 1 s than it does to stop the same car in 25 s.

To find the force involved in such a stop, you can use the following formula:

$$\text{force} = \frac{\text{mass} \times \text{change in velocity}}{\text{time}}$$

Sample Problem

What force is involved in stopping a 1200 kg car travelling at 100 km/h (28 m/s) in 25 seconds?

Steps in Solving It

1. Write down what you know.

mass = 1200 kg

change in velocity = final velocity (velocity₂) - starting velocity (velocity₁)

= 0 m/s - 28 m/s

= -28 m/s (the negative means the vehicle is slowing down)

time = 25 s

2. What do you want to find out?

What force is involved?

3. Use the formula.

$$\text{force} = \frac{\text{mass} \times \text{change in velocity}}{\text{time}}$$

$$= \frac{(1200 \text{ kg}) \times (-28 \text{ m/s})}{25 \text{ s}}$$

$$= \frac{-33\,600 \text{ kg}\cdot\text{m/s}}{25 \text{ s}}$$

$$= -1344 \text{ kg}\cdot\text{m/s}^2$$

4. State your findings.

The car experienced a negative or braking force of 1344 kg•m/s². **Note:** This is Car A in Figure 15.7 (on page 277).

Solve These

Use the example above to help you solve the following problems. Be sure to show all your calculations.

1. (a) During a race, a 1200 kg car slowed down to 100 km/h (28 m/s) just before it hit a brick wall. It stopped in 1 s. What force was involved? **Note:** This is Car B in Figure 15.7 (on page 277).
(b) How much larger is this force than the force in the sample problem?
2. A sport utility vehicle with a mass of 2000 kg is travelling at 100 km/h. The driver brakes to avoid a small animal in the road. The velocity changes from 28 m/s (100 km/h) to 8 m/s (30 km/h) in only 2 seconds. What force does the vehicle experience?

Check Your Understanding

1. How does the change in momentum compare between a quick stop (collision) and a controlled stop (braking) from the same velocity? Why?
2. In a collision that stops a vehicle, how does the original velocity relate to the force experienced by vehicle and passenger? Explain your answer using the terms “impulse” and “time.”
3. If you lengthen the time a change in velocity takes, will the force go up or down? Use the equation $\text{force} = \frac{\text{mass} \times \text{change in velocity}}{\text{time}}$ to support your response. Substitute some reasonable numbers for mass, velocity, and time to prove the effect.

15.4 Is Momentum Lost?

When an object slows down, momentum appears to be lost. Where does it go?

The energy of movement is changed to other kinds of energy.

- Friction in the brakes, and between the tires and the road, causes heat. In some crashes, this heat is enough to ignite the vehicle.
- Energy may also be lost in crumpling metal and shattering glass.

In many collisions, momentum is *not* all lost. Objects that collide continue to move in some direction. The amount of momentum lost will vary with the conditions and circumstances.

Conservation of Momentum

Imagine two vehicles travelling along a road in the same direction. Each has its own momentum. Since people usually refer to car velocities in km/h, we will use these units when calculating vehicles' momentum.

Now picture the vehicle behind rear-ending the vehicle in front. What happens to the momentum of each vehicle?

- The momentum of the vehicle in front will increase as it is pushed forward. It experiences a positive force.
- The momentum of the vehicle behind will decrease as the vehicle slows down because of the impact. It experiences a negative force.

Scientists have calculated the momentum of vehicles before and after a rear-end collision. Their calculations show that, under ideal conditions where friction is minimal and the vehicles lock together, the total momentum between the two vehicles does not change. This is also true if one vehicle bumps into another and there is no damage. During such a collision, one vehicle will lose momentum at the same rate that the other gains it.

You have heard of the Law of Conservation of Energy and the Law of Conservation of Mass. Momentum is also conserved. **The Law of Conservation of Momentum** means that the total momentum of two colliding objects stays the same after their collision.

READING

check

What happens to the energy of movement when momentum is lost?

READING

check

Use a rear-end collision to explain the Law of Conservation of Momentum.

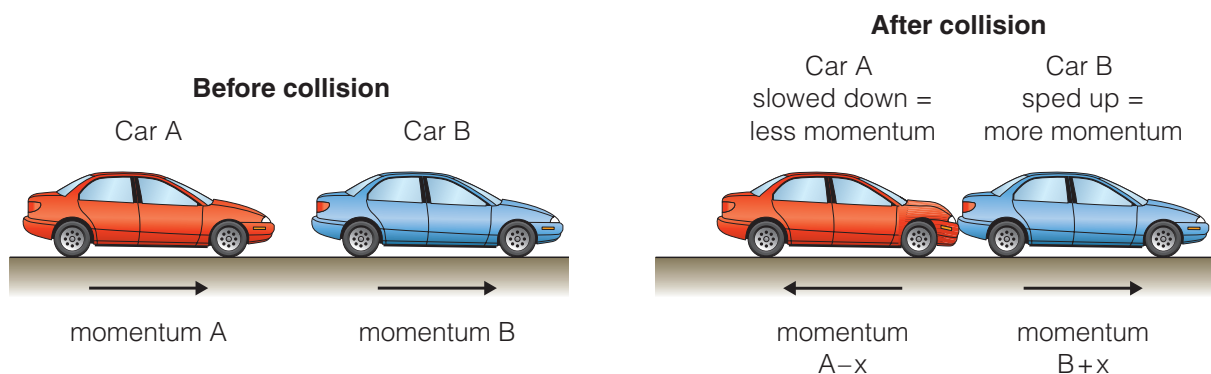


Figure 15.8 This rear-ender occurred on an icy road (therefore minimal friction) and locked the vehicles together. Imagine the momentum of each vehicle before and after the collision. Before the collision, Cars A and B were travelling at about the same velocity along the highway and had close to the same momentum. After the collision, the momentum of Car A has decreased — hitting Car B caused a negative impulse; this is the “ $-x$ ” in the equation on the left. The momentum of Car B has increased — being hit by Car A caused a positive impulse; this is “ $+x$ ” in the equation on the right.

Notice that the momentum of one vehicle *decreased* by the same amount that the momentum of the other vehicle *increased*.

Head-on Collisions

How does the Law of Conservation of Momentum apply to head-on collisions? Consider this.

Two vehicles are travelling along a highway in opposite directions. They have the same momentum. They hit head on. What happens?

Both objects are in motion and have the same momentum. The motion is in the opposite direction. The forces would cancel each other out. Their **total momentum** would be zero, and so would their velocity. They would stop completely (with a lot of crumpling and injury to the people inside).

If one vehicle had greater momentum than the other, that vehicle would add momentum to the other vehicle, which would then move backward.

READING
Check ✓

How does a head-on collision illustrate the Law of Conservation of Momentum?

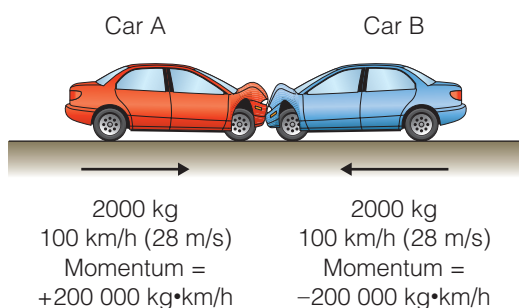


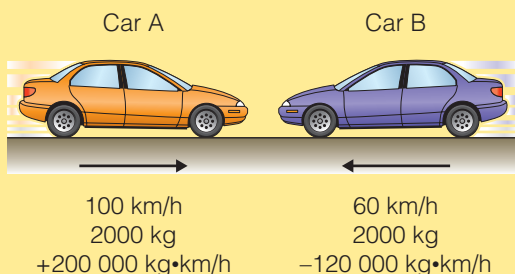
Figure 15.9 Cars A and B have the same mass and velocity. As a result, they have the same momentum. Note that the momentum of Car A has a positive sign (+) in front of it because Car A is moving to the right. Car B is moving to the left, so a negative sign (−) is used to indicate this opposite momentum.

What happens when the two cars collide? The value of their momentums will cancel each other out ($-200\,000\text{ kg}\cdot\text{km/h} + 200\,000\text{ kg}\cdot\text{km/h} = 0$). They will stop.

Hitting another vehicle head-on is like hitting a brick wall at the combined velocities of the two vehicles. Although the momentums cancel each other out, the force is enough to kill.

Calculating Final Momentum

Use the information about each collision to calculate the final momentum of the vehicles involved.



Sample Problem

Two cars with the same mass are travelling at different velocities. Car A is travelling at 100 km/h to the right. Car B is travelling at 60 km/h to the left. Car A has a momentum of +200 000 kg•km/h. Car B has a momentum of -120 000 kg•km/h.

- Calculate their total momentum.
- Which way will the vehicles move during a collision?

Steps in Solving It

- Draw what is happening.
- Write down what you know.
 momentum of car A = +200 000 kg•km/h
 momentum of car B = -120 000 kg•km/h

- What do you want to find out?

The total momentum.

- Add the two momentum figures.

$$+200\,000\text{ kg}\cdot\text{km/h} + -120\,000\text{ kg}\cdot\text{km/h} = +80\,000\text{ kg}\cdot\text{km/h}$$

- State your findings.

Both cars will move to the right. (Keep in mind that the bodies inside the cars continue to move with their original velocities and directions, unless restrained by seat belts or other safety devices.)

Solve This

Use the example above to help you solve the problem that follows. Draw a diagram. Show all your thinking and calculations.

- A transport truck with a mass of 10 000 kg and a car with a mass of 2000 kg are travelling at the same velocity (100 km/h) but in opposite directions. The truck is travelling to the left, and has a momentum of -1 000 000 kg•km/h. The car is moving to the right, and has a momentum of +200 000 kg•km/h. What is the total momentum after the collision? In which direction will these vehicles move if they collide?

Check Your Understanding

- Use a series of diagrams to illustrate the Law of Conservation of Momentum. Include a calculation showing the total momentum.
- Two vehicles are moving toward each other. The westbound vehicle has a momentum of 500 000 kg•km/h, while the eastbound vehicle has a momentum of 250 000 kg•km/h. If the two vehicles collide without braking, which way will they continue moving after the collision? Draw a diagram to illustrate your answer.

Key Terms

Law of Conservation of Momentum
total momentum

Getting Ready...

- Why are seat belts and air bags so effective in preventing injuries and fatalities?
- How is the seat belt designed?
- How can air bags be dangerous?
- Where and how should children sit in a vehicle, and why?



In the previous three chapters, you learned about how the laws of physics affect traffic safety and the motion of cars during collisions. This chapter is about using tests, like crash tests, to build safer cars.

Cars are safer today than they were even 10 years ago because safety engineers use what they learn from investigations with crash test dummies to build safer vehicles. Such work starts with research.

Researchers examine accident reports and identify safety needs. For example, research might show that passengers are slamming into dashboards on impact. Safety engineers use crash tests to help

them design a feature to prevent such problems.

Every part of the car is engineered with safety in mind.

In this chapter you will learn what safety features keep you safe. You will sit in the chair of a safety engineer as you consider how to improve these safety features.



16.1 How Safety Features Work

As a safety engineer, I create new safety features and improve old ones. To check how safe a vehicle is, we build human-like dummies and see what happens to them during collisions. We crash vehicle after vehicle until the safety feature we are developing works the way we want it to. Only then does the feature get installed in cars that will be sold to the public. Engineering processes take a lot of time and money – but they save lives.

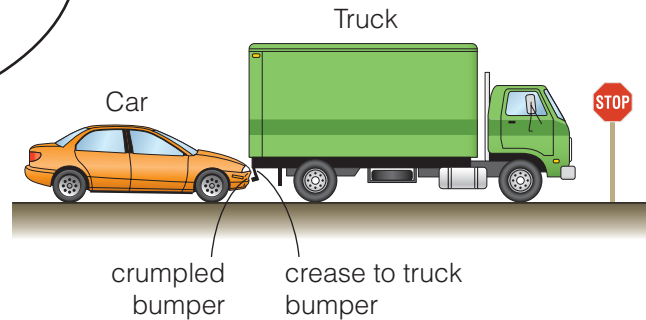


Figure 16.2 During accidents such as this, both bumpers crumple to absorb the force of the impact. This slows the force of the collision. Often, nobody is injured in such an accident but the bumpers need to be replaced.

Chapter 15 discusses the conservation of momentum. As explained in that chapter, when one object hits another, the momentum of the faster, heavier object is transferred to the slower, lighter object.

The occupants of vehicles experience various forces as their velocities change and momentum is transferred during a collision. The safety features on vehicles work to reduce the force of a crash. In Chapter 15 you learned that the force resulting from a collision is smaller if the change in momentum occurs over a longer period of time. This is the basic idea behind most vehicle safety features.

DidYouKnow?

An unrestrained child in a high-speed collision will have about the same momentum as a cannonball.

Safety features serve to slow down the collision. As a result, the occupants of the car experience less force and, therefore, receive fewer injuries.

Engineers use **crash test dummies** to determine what will happen to people during a collision. These are dolls that are as much like real human beings as modern technology can make them.

Engineers use these dummies to help them determine how well the **crumple zones** in cars work. **Crumple zones are parts of the car that are made of crushable metal or fibreglass. These usually include the hood, doors, and trunk. These sections crumple when the vehicle is hit. The crumple zones slow down the collision felt by the occupants and increase the time of the collision between the occupants and the interior of the vehicle, thus reducing the force experienced by the occupants.** This reduces the damage experienced by the occupants of the vehicle.

Types of Safety Features

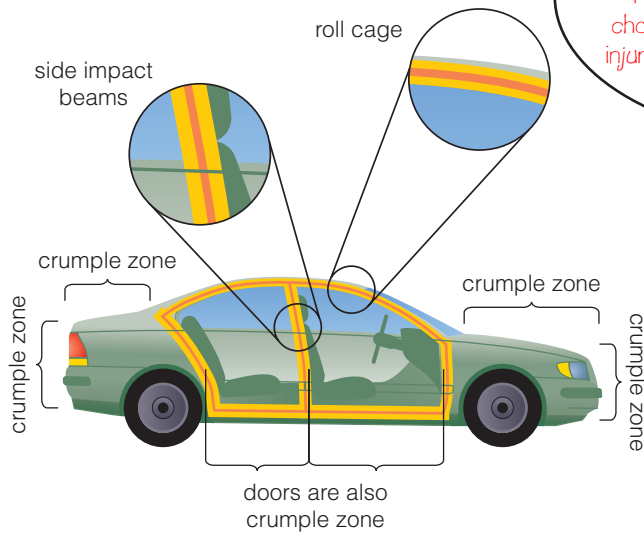


Figure 16.3 The rigid structure of the body of the car protects passengers. Notice how the rigid structure is protected by the various crumple zones.

Have you heard that it is better to be thrown clear of your vehicle during an accident? Don't believe it. Our research and tests show that your chances of survival and avoiding serious injury are much better if you are properly held in a safe position.



Figure 16.4 Safety engineers use tests to find out what kind of damage is done during accidents. Some of their findings may be surprising.

Crumple zones work on the outside of your vehicle. What else keeps you safe? As you probably noticed during the Starting Point activity, there are many different features. These include restraining features, operational features, and structural features.

- **Restraining features** hold occupants in one place. They prevent them from hitting the dashboard, windshield, sides, or roof of the car. They also prevent someone from being thrown out of the vehicle. Restraining features include seat belts, air bags, headrests, and child seats.
- **Operational features** operate all the time to keep you safe while driving. For example, headlights allow the driver to see clearly and warn other drivers of a vehicle's approach. Rear- and side-view mirrors assist the driver in seeing all parts of the road. Tail and signal lights warn other drivers of changes in speed and direction. Brakes allow the vehicle to slow safely.
- **There are two types of structural features: rigid features and crumple zones.**
 - Rigid features include the heavy frame, a roll cage, and side-impact beams in the doors. Each of these is built from materials that will not crumple or break under extreme pressure. In this way, they are designed to protect the people inside from anything that may hit the car.
 - The crumple zones include the hood, trunk, and fenders. These may be completely crushed in a bad accident. If the rigid frame stays intact, however, the cabin of the vehicle will keep the occupants safe.

READING
check ✓

What is the purpose of each type of safety feature? Give an example of each type.

READING check

Why are aluminum and plastics used frequently in the bodies of vehicles?

Safety Features at Work

Study Figures 16.5 to 16.8 and read the captions. Identify the type of feature being discussed and how it works.

Figure 16.5 1967 Dodge Dart



I loved the sleek lines of my father's '67 Dodge Dart. Those heavy old cars had a lot of momentum when they got going! Unfortunately, in a collision, the heavy metal body of cars like this one did not absorb any of the impact. Whiplash was the least of the worries of the people inside. Many were seriously injured when they hit the hard dashboard or the rigid steering wheel.



Have you seen the results of our latest tests? The vehicles that crumpled absorbed a lot of force during a collision. That means less damage was done to our crash test dummies.

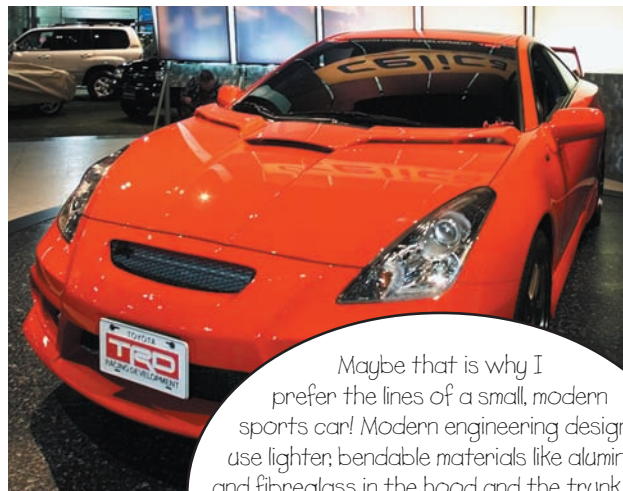


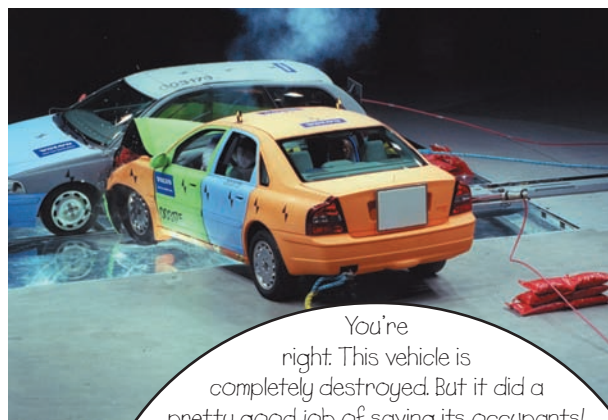
Figure 16.6

Maybe that is why I prefer the lines of a small, modern sports car! Modern engineering designs use lighter, bendable materials like aluminum and fiberglass in the hood and the trunk. The hood and trunk look as good as the body on those old Darts, but the ones on the new cars crumple in a collision. That's hard on the car, but protects the people inside.

Figure 16.8 Modern cars absorb much of the force of an impact by crushing. This reduces the force of a collision and protects the occupants.



Figure 16.7 Identify the crumple zones on this vehicle.



You're right. This vehicle is completely destroyed. But it did a pretty good job of saving its occupants! The outer panels crumpled, but they protected the cabin of the vehicle. That is what I have been working on for the past 10 or so years. We built and tested better roll cages. By making them of rigid materials, we formed a protective cage around the occupants. Later, we added side-impact beams to protect occupants from collisions from the side. Then we added a beam through the dash so the steering column and engine could not crush their way into the cabin.





Figure 16.9 Imagine the amount of force behind a truck hitting a concrete barrier such as this one.



Figure 16.10 Now picture a similar accident — only this time, the truck hits a barrier like the one shown here. How might a crash cushion protect a vehicle and its occupants?

Road Safety Features

Vehicles are not the only things with safety features. Many roads also have **road safety features** that reduce the force of a collision by slowing the impulse.

One such technology is the **crash barrier or crash cushion**. Barrels filled with water (or sand) crush or burst on impact. This applies a slowing force to the vehicle without doing as much damage as a solid barrier.

Today's metal **guardrails** will bend (but not break) to lower the momentum of a vehicle and absorb the force of collision. In this way, guardrails prevent the vehicle from rolling over or rebounding into traffic. Some guardrails are designed so the entire section will snap off its posts if a vehicle hits any section of the railing head on. This serves to slow the vehicle safely by applying a consistent medium strength force.

Light poles used to be immovable. Today's light poles are designed with shear pins that snap off under the force of collision. The vehicle hits the light post, but is not immediately stopped. This reduces the force of impact.

READING Check

Describe how one feature of the road is designed to lower the force of collision.

Check Your Understanding

1. Why are cars with crumple zones better than cars that do not crumple?
2. What parts of modern vehicles are meant to be rigid? Why?
3. Describe how two road features were redesigned to reduce the force of impact.
4. Why is it important to regularly maintain the safety features of any vehicle? Mention two safety features that should be checked regularly, and explain why this is important.

Key Terms

crash test dummies
crumple zones
restraining features
operational features
structural features
road safety features
crash cushion
guardrails

16.2 Seat Belts

READING Check

Explain the difference between how a lap belt and a shoulder harness work. Which one provides better protection?

A lot of work has been done to develop a seat belt that is safe *and* comfortable. Modern seat belts are designed to stop your body from being thrown forward during a collision. The original designs were **2-point belts or lap belts**. These held a person across the pelvis. Unfortunately, that did not stop the upper body from moving.

The modern **3-point belt or shoulder harness** was designed to help restrain the upper body. Now that the entire body is restrained, it is only the head you have to worry about.

So far, safety engineers have not developed a way to restrain a person's head during a collision. That is why it is important to use a raised headrest along with the seat belt. The importance of properly adjusted headrests and what happens if you do not use them was discussed in Chapter 13.

Find Out **ACTIVITY**

Seat Belts Then and Now

How has the structure of seat belts changed since seat belts were first developed?

What to Do

1. Compare the different features in the seat belts in Figures A and B, and the one in Figure 16.13 on page 293.
2. The figures are shown chronologically. Examine the figures and consider what design changes have occurred.



A. 2-point belt



B. 3-point belt

SKILL CHECK

Initiating and Planning

Performing and Recording

Analyzing and Interpreting

Communication and Teamwork

How Seat Belts Work

Have you ever been hit by something that flew off the back window ledge or seat when a car stopped fast? Then you have some idea about what could happen if you were not wearing your seat belt during a collision. Your body would continue travelling at the same velocity the car was going before the collision.

Modern seat belts have been carefully engineered to make sure they are there, and working, when you need them. Consider the technology that goes into making them safe.

Webbing — Seat belt material is made up of thousands of thin strands of polyester or nylon. **The arrangement of these fibres allows the belt to stretch from 10 to 15 percent of its normal length during a collision. This stretching allows the passenger to stop more gradually and feel a smaller force.**

Retractor unit — Inside the retractor portion of the seat belt is a clever system that involves a pendulum and a ratchet. The ratchet maintains tension in the webbing. This is why the belt is always snug on your body. By pulling gently on the webbing, you can reel out more of the belt. However, if the vehicle experiences a sudden stop, the pendulum in the ratchet mechanism swings forward. This locks a bar in place that prevents more webbing from coming out.

READING check

Why would we want a seat belt to stretch a small amount in a collision?



Figure 16.11 What regular maintenance would this part of the seat belt likely need?

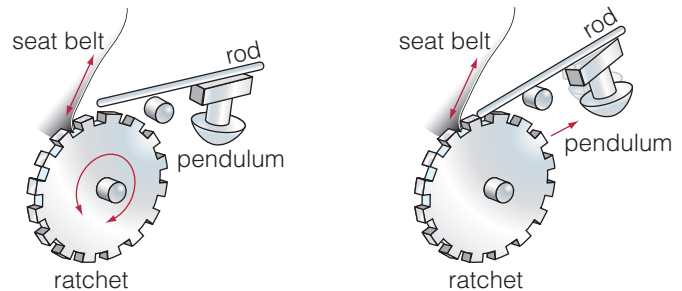


Figure 16.12 In most seat belt systems, the belt webbing is attached to a retractor mechanism that uses a spring and spool to keep the webbing snug. A second mechanism comes into play in a collision. The ratchet and pendulum system shown here causes the seat belt spool to lock when the car decelerates quickly. How does this technology keep you comfortable and also safe?



Figure 16.13 4-point belts. These belts restrain both shoulders and distribute the force over more of the chest. This reduces stress on the ribs, heart, and lungs.

Child Restraints

The Canadian Automobile Association reports that up to one third of Canadian children are not properly restrained while in a motor vehicle. This leads to as many as 10 000 injuries per year.

Children have weaker bones and muscles along with their comparatively heavy heads. Devices for children are designed to provide the extra protection this type of body needs.

Types of child seats differ based on the mass of the child. The back seat of the car is recommended for children because there is no dashboard to hit and no chance of injury from an air bag. Read the captions in Figures 16.17 to 16.20 to learn about the different kinds of children's seat and when they are used.

READING check

List two reasons why children need different restraining devices than adults.



Figure 16.15 Infant-only seats are for children who cannot sit up by themselves yet. These are designed to face the back of the vehicle. The force of a crash would be absorbed by the entire back of the child.



Figure 16.16 Children who weigh more than 9 kg usually have bones and muscles strong enough for a **child seat**. In this kind of seat, the child faces forward. It is still best to have such seats in the back seat.



Figure 16.17 Convertible child seats are used with children from birth to a mass of 18 kg. They can be used with infants and children.



Figure 16.18 Booster seats are made for older children who are not yet tall enough to use the seat belt properly without being in a raised position.

Check Your Understanding

1. What advantages does a 3-point belt have over a lap belt in a collision?
2. Why does the pendulum in a seat belt swing forward when the vehicle stops suddenly?
3. Describe briefly how a seat belt should be worn. Sketch a diagram and list three important points.
4. (a) Why shouldn't a baby be carried in a passenger's lap? Describe some of the things that could happen in a collision.

Key Terms

2-point belts
lap belts
3-point belt
shoulder harness
infant-only seats
child seat
convertible child seats
booster seats

16.3 The Evolving Air Bag

DidYouKnow?

First generation air bags deployed so fast that they hit the driver at over 300 km/h!



Figure 16.19 When inflated, air bags fill the front part of a vehicle. The air inside cushions the occupants and slows their velocity and momentum.



Figure 16.20 The systems in Figure 16.21 and this photo both use air to reduce momentum.

READING check

How do air bags work?

An **air bag** is rolled up inside the steering wheel and the passenger-side dashboard. When activated, it performs much the same function as a parachute. Parachutes protect skydivers by safely lowering the velocity and momentum of their fall. Similarly, **air bags safely slow the velocity and momentum of the occupant of a car.**

In a collision, **a chemical reaction inside the air bag causes it to inflate in 30 milliseconds.** Sensors in the car send an electric signal to ignite a canister of a chemical called sodium azide ($\text{NaN}_3(\text{s})$). The chemical quickly decomposes to release nitrogen gas ($\text{N}_2(\text{g})$). A small amount (130 grams) of sodium azide crystals will produce 67 litres of nitrogen gas — enough to inflate a normal air bag. When a car occupant hits the bag, the bag reacts in the same way as the crash cushions on page 291 to cushion the force resulting from the collision.

An air bag adds an extra cushion to help the seat belt restrain occupants of the front seat. The first air bags were designed with adult males in mind. These are called **first generation air bags**. These first air bags inflated too much and too quickly. This caused injuries to smaller people, especially small children. Some people did not wear seat belts because they thought they did not have to when they had an air bag. Some of these people died from colliding with first generation air bags, which were too hard!

We spent a lot of time working on the **second generation design** to improve the air bags. Air bags should cushion a passenger softly. If the bag does not inflate enough, the passenger may get whiplash or hit the steering wheel or dashboard. If the air bag is too full, it acts like a brick wall. We did a lot of tests to help us figure out how much an air bag should be inflated.

DidYouKnow?

Do not place small children in the front passenger seat of a vehicle when the air bag is activated. Small children have been killed as a result of the force of collision with the air bag during even minor crashes.



Safety Systems Continue to Improve

Safety engineers continue to improve the safety features of vehicles. The first generation air bags caused many problems. After trials, testing, and improvement, the second generation design worked better. Second generation air bags deploy at a safer speed, at a safer force (from 20 to 35 percent less than the first generation ones), and work with the seat belt to protect car occupants.

Newer, or **third generation air bags**, are even better than the ones that came before. There are now seat sensors that detect the mass of the occupant and adjust the force of the air bag accordingly. Some of the new air bags also have switches so they can be turned off if there is a child in the front seat.

Modern seats are now designed to act like a baseball catcher's mitt. They change shape to cradle a body that is thrown against them.

- Belts and air bags should catch you gently but firmly as you fly forward.
- Seats and raised headrests should catch you gently but firmly as you fly back.

If the safety design features are all working and adjusted correctly, it is possible to come through a serious crash practically without injury!

DidYouKnow?

Statistics show that air bags have saved about 30 times the number of people who have been killed by them. Most of the people injured or killed by air bags were not wearing seat belts.



How are modern air bags safer than the original ones?

DidYouKnow?

Crumple zones occur in the interior of modern cars as well. The steering wheels and dashboards of newer cars are soft and pliable to reduce the force of impact.

Check Your Understanding

1. What can an air bag do that a seat belt cannot do?
2. List several reasons why you should not depend on the air bag alone.
3. How can air bags injure children?
4. Why was it necessary to reduce the force of the first generation air bags?

Key Terms

air bag
first generation
air bags
second generation
design
third generation
air bags